Local LLM Reliability for Ideal Rocket Nozzles

Ideal Rocket-Nozzle Performance: A Preliminary Evidence and Physics Audit

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Research question

Does technical retrieval help a small local LLM produce physically valid nozzle answers?

The comparison checks values, units and applicability, with a separate audit of evidence IDs.

Same model and question inputs in both conditions. An independent evaluator checks the outputs.

PRELIMINARY: 95 saved responses, 18 completed pairs. Planned full study: 216 responses.

Why this topic fits rocket propulsion

Thrust connects mass flow, exhaust speed and exit pressure to an interpretable propulsion quantity.

Ideal nozzle equations make a bounded numerical reference possible.

A local model and a short fixed corpus make the comparison feasible on the existing CPU setup.

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Motivation and significance

A plausible number can hide a missing pressure-thrust term or a wrong unit conversion.

Retrieving an equation does not ensure correct arithmetic or a valid flow assumption.

Retaining wrong answers makes the failure observable and useful for engineering education.

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Related research

Study	Approach	Relevance
Yang et al., 2026	Combustion knowledge injection	Domain retrieval and evaluation
ThermoQA, 2026	Programmatic thermo references	Numerical reasoning benchmark
PhyNiKCE, 2026	Physics constraints in CFD agents	Explicit physical validation
ChatCFD, 2025 / 2026	Structured CFD automation	Nozzle precedent already exists

A scoped review. No first-ever LLM or nozzle claim.

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The bounded research gap

Within the reviewed studies, this joint local-model evaluation remains insufficiently characterised.

The focus combines nozzle values, dimensions, applicability, missing inputs and evidence traceability.

The contribution is a controlled, inspectable measurement for one quantized local model.

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Governing physical relations

Quantity	Ideal relation
Axial thrust	F = mdot Ve + (pe - pa) Ae
Exit temperature	Te = T0 / [1 + (gamma - 1) Me^2 / 2]
Exit speed	Ve = Me sqrt(gamma R Te)
Choked mass flow	mdot = p0 At / sqrt(T0) sqrt(gamma/R) [2/(gamma+1)]^[(gamma+1)/(2(gamma-1))]

Fixed gas properties and a stated choking / branch assumption are required.

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Assumptions and validity limits

Included in the reference	Outside this reference
Steady one-dimensional ideal gas	Transient and multidimensional flow
Constant gamma and composition	Dissociation and variable specific heats
Sonic throat, shock-free supersonic branch	Unchoked throat and internal shock
Momentum and pressure thrust	Viscous loss and nozzle separation

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Worked ideal-nozzle example

Input	Computed reference
p0 = 1 MPa, T0 = 900 K	Me = 2, Te = 500 K
At = 0.001 m^2, Ae/At = 1.6875	Ve = 896.44 m/s
gamma = 1.4, R = 287 J/(kg K)	pe = 127.805 kPa, mdot = 1.347281 kg/s
pa = 100 kPa	Total thrust = 1254.67 N

The pressure contribution is included in total thrust.

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Reference verification

35 supersonic area-Mach roots: bisection cross-checked with SciPy brentq.

Analytic identity: gamma = 1.4 and Me = 2 give Ae/At = 1.6875.

Mass continuity and an independent energy relation provide further checks.

Current local tests: 106 passed. Reference checks: passed.

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Controlled answering experiment

Element	Frozen choice
Model	Qwen3.5-4B Q4_K_M, llama.cpp CPU
Condition A	Same question, no supplied passages
Condition B	Same question, BM25 top three passages
Evaluation	Independent reference after saving the response

Same quantization, generation settings and matched seeds in both conditions.

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Frozen benchmark

Family	Evaluation purpose	Evaluation cases
Choked mass flow	Stagnation inputs and sonic throat	6
Thrust	Momentum and pressure terms	6
Exit speed	Supersonic area-Mach branch	6
Mixed units	Conversions before calculation	6
Applicability / missing inputs	Correct abstention in two families	12

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Evaluation criteria

Criterion	Frozen rule
Combined correctness	Valid JSON, correct decision and issue, correct converted value
Numerical tolerance	max(atol, 1% of absolute reference)
Units	Pint dimension check and conversion to reference SI unit
Evidence audit	Retrieved reference recall and cited-ID precision
Failure handling	Wrong / malformed answers remain incorrect

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Reproducible execution

The frozen source and benchmark checkpoint precedes model inference.

Bounded nonroot Docker jobs use a private internal network and expose no host API port.

Raw requests, answers, scores and plots persist in resutls/. Operational records persist in logs/.

Interrupted compatible runs resume. Every saved checkpoint triggers a normal Git push attempt.

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Preliminary paired reliability

Metric	LLM only	BM25 RAG
Correct / paired records	5 / 18	6 / 18
Combined accuracy	27.8%	33.3%
Schema-valid responses	14	16
Median request wall time	71.8 s	76.3 s

18 matched case/seed pairs. RAG difference: +5.6 percentage points.

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Preliminary results by question family

Family	LLM only	BM25 RAG
Mass flow	0/4 (0.0%)	0/4 (0.0%)
Thrust	0/2 (0.0%)	0/2 (0.0%)
Exit speed	0/3 (0.0%)	0/3 (0.0%)
Mixed units	0/3 (0.0%)	0/3 (0.0%)
Applicability	0/1 (0.0%)	1/1 (100.0%)
Missing inputs	5/5 (100.0%)	5/5 (100.0%)

Complete pairs only. N/A means no completed pair for that family.

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Paired outcomes and failure audit

Paired outcome / error	Count
Both conditions correct	5
RAG alone correct / baseline alone correct	1 / 0
Both conditions incorrect	12
Schema failures, baseline / RAG	4 / 2
Numerical or unit failures, baseline / RAG	9 / 10

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One recorded paired example

A steady, shock-free, calorically perfect ideal nozzle has a sonic throat and a supersonic exit. Ae/At=2.2, chamber stagnation T0=1250 K, gamma=1.4, specific R=330 J/(kg*K). Calculate exit speed in m/s using the supersonic area-Mach branch.

Independent target: 1219.26 m/s

LLM only: 2967.8 m/s, decision answer. Score: numeric failure.

BM25 RAG: Invalid JSON.
{"decision":"answer","value":3680.5,"unit":"m/s","issue":"none","evidence_ids":[],....
Score: schema failure.

exit_velocity-04, seed 17. Short excerpts. Full question and raw responses are retained.

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Expected and observed outcomes

Before measurement	After this run
Retrieval may improve equation access	RAG difference: +5.6 percentage points
Arithmetic or applicability errors may remain	Incorrect responses: 13 baseline, 12 RAG
Zero or negative RAG benefit is plausible	No minimum accuracy gain was promised
Planned full coverage: 216 responses	95/216 saved; full study unfinished

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Potential impact and limitations

The result is an inspectable audit connecting a local LLM to explicit nozzle physics.

It separates formatting, numerical, dimensional and applicability failures.

Limits: one model, correlated original templates, a short corpus and an ideal physical domain.

Next work: expert review and a controlled test of a calculator or input validator.

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Demonstration and reproduction

Show the saved Git checkpoints and docs/final/FINAL_MANIFEST.json.

Explain worked-example.json and its ideal-flow assumptions.

Open study-report.html and compare the paired raw answers.

Use scripts/complete_project.py publish to retry a pending Git push.

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Appendix: full protocol

Protocol item	Recorded setting
Evaluation schedule	36 cases x 2 conditions x 3 seeds = 216
Seeds and generation	17, 43, 89. Temperature 0.7, top-p 0.8, top-k 20
Output / context budget	768 output tokens, 4096 context, one request at a time
Retrieval	Lucene BM25, k1=1.2, b=0.75, top-k=3
Human review	Physical reasoning and claim support pending

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Appendix: area-Mach derivation

$$\frac{A_e}{A_t} = \frac{1}{M_e}\left[\frac{2}{\gamma+1}\left(1+\frac{(\gamma-1)M_e^2}{2}\right)\right]^{\frac{\gamma+1}{2(\gamma-1)}}$$

At $\gamma = 1.4$ and $M_e = 2$:

$$\frac{A_e}{A_t} = \frac{1}{2}(1.5)^3 = 1.6875$$

$$\frac{T_e}{T_0} = \frac{1}{1.8} = \frac{5}{9}$$

References: related research

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References: physics and implementation

[5] NASA Glenn. Thrust Equations Summary
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[6] NASA Glenn. Compressible Area Ratio
www.grc.nasa.gov/www/k-12/rocket/astar.html

[7] BM25S implementation
github.com/xhluca/bm25s

[8] NIST. Guide for the Use of SI, SP 811
www.nist.gov/pml/special-publication-811